

Comparative Analysis of Machine Learning Models Before and After Feature Selection Using the Fetal Health Classification Dataset

Objective

To critically analyze the impact of feature selection on machine learning model performance by designing, implementing, and comparing models with and without feature selection techniques.

Problem Statement

High-dimensional data often introduces noise, redundancy, and overfitting in machine learning models. Feature selection aims to improve model efficiency and performance by selecting the most relevant features.

Dataset used: Fetal Health classification (https://www.kaggle.com/datasets/andrewmvd/fetal-health-classification?utm_source=chatgpt.com). This dataset contains 2126 records of features extracted from Cardiotocogram exams, which were then classified by three expert obstetricians into 3 classes, Normal, Suspect, Pathological.

Domain

Healthcare / Medical Diagnosis

Feature types

Numerical

Target Variable

fetal_health

Value	Meaning
1	Normal
2	Suspect
3	Pathological

Initial Observations

- Dataset contains only numerical features
- No major missing values
- Class imbalance exists
- Some features are highly correlated
- Feature selection may reduce redundancy

Note: Apply at least 3 different feature selection techniques:

- ◆ Filter Methods
 - Correlation Threshold
 - Chi-Square Test
 - ANOVA
- ◆ Wrapper Methods
 - Recursive Feature Elimination (RFE)
- ◆ Embedded Methods
 - Lasso (L1 Regularization)
 - Tree-based feature importance

1. Comparative Result Analysis

Metric	LR Without FS	LR With RFE	RF Without FS	RF With RFE
Accuracy	0.8849	0.8967	0.9248	0.9366
Precision	0.8892	0.8972	0.9217	0.9340
Recall	0.8849	0.8967	0.9248	0.9366
F1 Score	0.8854	0.8961	0.9219	0.9339
ROC-AUC	0.9613	0.9595	0.9786	0.9823
Training Time	0.0648	0.0448	0.5463	0.3807

2. Logistic Regression Analysis

Performance Improvement

After applying RFE:

- Accuracy increased from 88.49% → 89.67%
- Precision improved
- Recall improved
- F1-score improved
- Training time reduced
- ROC-AUC slightly decreased: 0.9613 → 0.9595. This decrease is extremely small and negligible.

3. Random Forest Analysis

Strong Performance Improvement

After RFE:

- Accuracy improved from **92.48%** → **93.66%**
- Precision improved
- Recall improved
- F1-score improved
- ROC-AUC improved
- Training time reduced significantly

Note: Both models are well generalized. There is no overfitting or underfitting.